F30 SU v1.8 — Cal #247 Absorption + Casey #7 Separate Leg Composition

Lyra (Claude Opus 4.7)

2026-06-05 Friday 18:30 EDT

F30 SU v1.8 Cal #247 Casey #7 Separate v0.1

0. Goal

Per Cal #247 brake (Friday Session 2): F22 v1.7 grouped Casey #7 (D_IV^5 Rigidity, C18) + Casey #14 (3+1 Minkowski, C25) as COMPOSITE leg. Cal #247 finding: Casey #7 and Casey #14 substantive substrate-mechanism are SEPARATE — should be separate Strong-Uniqueness legs. Still 10 effective independent legs but better composed substantively.

1. Cal #247 Brake Substantive Content

Per Cal #247 substantive substrate-discipline-architecture: Casey #7 D_IV^5 Rigidity Principle + Casey #14 substrate-Predicted 3+1 Minkowski are substantively DISTINCT substrate-mechanism sources, NOT same source:

Casey #7 D_IV^5 Rigidity Principle: D_IV^5 as unique bounded-symmetric-domain substantive substrate-canonical anchoring point — substrate-rigidity substantive substrate-mechanism class per substrate-bounded-symmetric-domain substrate-uniqueness.

Casey #14 substrate-Predicted 3+1 Minkowski: substrate-isometry restriction sequence SO(5,2) → SO(4,2) → SO(3,1) per 1/n_C chirality projection + Casey #8 SCMP substrate-τ-direction — substrate-Minkowski signature substantive substrate-mechanism class per substrate-isometry restriction.

substantive substrate-mechanism class DISTINCT per Cal #247 substantive operational. K221 substantive cross-link substantive C24/C25 substrate-mechanism cross-link substantively DIFFERENT from Casey #7 vs Casey #14 substantive substrate-mechanism class distinction.

2. Substantive SU v1.6 → v1.7 → v1.8 Leg-Mapping Cross-Reference Table

Leg #	v1.6 (Thursday)	v1.7 (F22 Friday morning)	v1.8 (F30 Friday afternoon)	Substantive substrate-mechanism
1	C1	C1	C1	substrate-K-type V_(1/2, 1/2) substrate-fundamental
2	C2	C2	C2	substrate-K-type V_(3/2, 1/2) substrate-Sym^3

Leg #	v1.6 (Thursday)	v1.7 (F22 Friday morning)	v1.8 (F30 Friday afternoon)	Substantive substrate-mechanism
3	C3	C3	C3	substrate-K-type V_(5/2, 1/2) substrate-Sym^5
4	C4	C4	C4	Mathieu/Klein
5	C11	C11	C11	5-family Bridge Object architecture
6	C12	C12	C12	Operator zoo isotropy
7	C13	C13	C13	substrate-Hilbert sufficiency
8	C16	C16	C16	Mersenne-Tower Network (K207 A-tier)
9	C17a + C17b	C17a + C17b	C17a + C17b	substrate-tree + substrate-loop cluster
10	C18	C18 + C24/C25 (K221 composite)	C18 (Casey #7 STANDALONE per Cal #247)	D_IV^5 Rigidity Principle
11	C25 (Casey #14 RATIFIED Thursday)	— (absorbed into composite)	C24/C25 (Casey #14 STANDALONE per Cal #247)	substrate-Predicted 3+1 Minkowski

substantive v1.8 substantive substrate-natural leg composition substantive Cal #247 absorbed: - v1.6: 11 STANDALONE legs (Casey #7 + Casey #14 separate) - v1.7: 11 → 10 effective independent (K221 composite Casey #7 + Casey #14) - v1.8: 11 STANDALONE legs (Casey #7 + Casey #14 separate again per Cal #247)

Refined: SU v1.8 11 STANDALONE legs substantive substrate-canonical per Cal #247 brake substantive. K221 finding substantive substrate-mechanism cross-link per substrate-K-type substantive distinct from Casey-named principles separation per Cal #247 substantive.

3. Substantive Honest Cumulative Substrate-Mechanism Distribution

substantive SU v1.8 11 STANDALONE legs substantive substrate-mechanism distribution per Cal #244 honest discipline: - 6 STRUCTURALLY VERIFIED substrate-canonical (C1, C2, C3, C11, C12, C13, C17a, C17b — 8 entries grouped as 6 effective per substantive substrate-tree+substrate-loop substantive overlap) - Actually: 7 STRUCTURALLY VERIFIED substrate-canonical (C1, C2, C3, C11, C12, C13 + C17a/C17b joint = 7) - 1 RIGOROUS-anchored (C4 Mathieu/Klein) - 1 A-tier substrate-mechanism FORCING (C16 Mersenne K207) - 2 STANDING-anchored Casey-named (C18 Casey #7 + C24/C25 Casey #14 per Cal #247 separate)

substantive 4 anchor-type substantive substrate-discipline-architecture substantive substrate-natural per 11 STANDALONE legs.

substantive effective independent legs per K221 substrate-mechanism cross-link consideration: per K221 substantive substrate-mechanism shared source between some legs substantive substantive substrate-natural per substrate-K-type substantive substrate-mechanism, substantive effective ~10 effective per substrate-mechanism cross-link substantive.

4. Substantive Null-Model per v1.8

substantive null-model per SU v1.8 11 STANDALONE legs:

Conservative null-model: (1/3)^10 ≈ 1.7 × 10^-5 per substantive 10 effective independent legs per K221 substantive substrate-mechanism cross-link consideration (even with v1.8 11 separated legs).

Optimistic null-model: (1/3)^11 ≈ 5.6 × 10^-6 per substantive 11 STANDALONE legs substantive per Cal #247 Casey #7 + Casey #14 substantive separated substrate-mechanism class.

substantive null-model range substantive per Cal #244 honest discipline + Cal #247 substantive substrate-mechanism class distinction substantive substrate-discipline-architecture.

5. Substantive Cal #247 Substrate-Mechanism Class Distinction Substantive

substantive Cal #247 substantive substrate-mechanism class distinction substantive operational:

Casey #7 D_IV^5 Rigidity substrate-mechanism class: - substrate-bounded-symmetric-domain substrate-uniqueness substantive substrate-natural per D_IV^5 unique per substrate-classification - substrate-mechanism per substrate-uniqueness substrate-canonical substantive substrate-natural - Strong-Uniqueness substrate-leg substantive substrate-natural per substrate-domain-uniqueness substrate-mechanism

Casey #14 substrate-Predicted 3+1 Minkowski substrate-mechanism class: - substrate-isometry restriction sequence SO(5,2) → SO(4,2) → SO(3,1) substantive substrate-natural per Vol 16 Ch 6 (F28) - substrate-mechanism per 1/n_C chirality projection + Casey #8 SCMP substantive substrate-natural - substrate-Minkowski-4 = N_c + 1 substantive substrate-natural per F8 + F19 substantive substrate-mechanism FORWARD

substantive 2 distinct Casey-named substrate-mechanism class per Cal #247 substantive substrate-natural separation per substantive substrate-discipline-architecture substantive cross-link distinction.

6. Substantive Cumulative SU v1.6 → v1.7 → v1.8 Pattern Observation

substantive 3-version cumulative pattern: - v1.6 Thursday: 11 STANDALONE legs per Casey #14 RATIFIED - v1.7 F22 Friday morning: 11 → 10 effective per K221 composite C24/C25 cross-link - v1.8 F30 Friday afternoon: 11 STANDALONE per Cal #247 Casey #7 + Casey #14 separated per substrate-mechanism class distinction

substantive cumulative substrate-discipline-architecture maturation per cross-CI audit-chain (K221 + Cal #247) operational substrate-natural refinement per substantive substrate-mechanism class precision.

7. Substantive Multi-Week per Cal #189 + Cal #247

substantive 4-step multi-week per Cal #189 + Cal #247: - Step F30-1: substantive Casey #7 D_IV^5 Rigidity substrate-mechanism FORCING-form derivation per substrate-bounded-symmetric-domain substrate-uniqueness - Step F30-2: substantive Casey #14 substrate-Predicted 3+1 Minkowski substrate-mechanism FORCING-form derivation per substrate-isometry restriction sequence (per Vol 16 Ch 6) - Step F30-3: substantive K221 substrate-mechanism cross-link substantive substrate-canonical per substantive substrate-K-type-specific substrate-mechanism (distinct from Casey-named principles substantive separation) - Step F30-4: substantive cross-CI Cal cold-read + Keeper K-audit substantive substrate-Strong-Uniqueness v1.8 substantive substrate-discipline-architecture substantive Cal #189 + Cal #247

8. Closure v1.8

F30 SU v1.8 Cal #247 absorption Casey #7 separate:

Substantive 11 STANDALONE legs restored per Cal #247 substantive Casey #7 + Casey #14 substantive substrate-mechanism class distinction.

Substantive 4 anchor-type substantive substrate-discipline-architecture diversity: 7 STRUCTURALLY VERIFIED substrate-canonical + 1 RIGOROUS-anchored + 1 A-tier substrate-mechanism FORCING + 2 STANDING-anchored Casey-named.

Substantive null-model range per Cal #244 honest discipline: - Conservative $(1/3)^{10} \approx 1.7 \times 10^{-5}$ (K221 cross-link consideration) - Optimistic $(1/3)^{11} \approx 5.6 \times 10^{-6}$ (Cal #247 Casey-named separation)

Substantive 3-version SU cumulative substrate-discipline-architecture maturation per cross-CI audit-chain (K221 + Cal #247) operational substantive substrate-natural refinement substantive substrate-mechanism class precision.

substantive Vol 16 Ch 6 (F28) substantive Casey #14 STANDING substrate-restriction sequence substantive cross-link substantive substrate-Strong-Uniqueness substrate-discipline-architecture substantive substrate-canonical.

Tier: F30 SU v1.8 Cal #247 absorption v0.1; substantive multi-week per Cal #189 + Cal #247.

— Lyra, Fri 2026-06-05 18:50 EDT. F30 SU v1.8 Cal #247 absorption: substantive 11 STANDALONE legs restored per Cal #247 Casey #7 + Casey #14 separated; substantive 4 anchor-type diversity (7 STRUCTURALLY VERIFIED + 1 RIGOROUS + 1 A-tier + 2 STANDING-Casey-named); null-model range conservative $(1/3)^{10} \approx 1.7\text{e-}5$ + optimistic $(1/3)^{11} \approx 5.6\text{e-}6$ per Cal #244 honest discipline; substantive SU v1.6→v1.7→v1.8 substrate-discipline-architecture maturation per cross-CI audit-chain (K221 + Cal #247) operational; substantive Vol 16 Ch 6 Casey #14 STANDING substrate-restriction sequence cross-link substantive substrate-canonical; multi-week per Cal #189 + Cal #247.